

Calculus and Linear Algebra

Rank and Linear Systems

Lecture 7

Daniele Angelini
Sapienza University of Rome

30 September 2026

Lecture outline

- ▶ Rank: minors, independent rows and columns.
- ▶ Gaussian elimination, pivots and echelon form.
- ▶ Compatible and incompatible systems.
- ▶ The Rouche'–Capelli theorem.
- ▶ Guided examples: 2×2 , 3×2 , 2×3 , 3×3 .

Today's aim

Decide whether a system has solutions, classify them, and solve systems with a unique solution.

Formal definition using minors

A **minor of order** k is the determinant of a square submatrix obtained by choosing k rows and k columns. **Definition.** The rank is the largest order of a non-zero minor.

$$\text{rank}(A) = r > 0$$

means that at least one minor of order r is non-zero and all minors of higher order vanish.

Bounds and the zero matrix

For $A \in \mathbb{R}^{m \times n}$, $0 \leq \text{rank}(A) \leq \min\{m, n\}$. The zero matrix has rank 0.

A practical test with consecutive orders

For $1 \leq r \leq \min\{m, n\}$, it is enough to check:

- ▶ at least one minor of order r is non-zero;
- ▶ every minor of order $r + 1$ is zero, when such minors exist.

Why this is enough

Laplace expansion shows that if all minors of one order vanish, then all minors of the next order vanish as well.

If $r = \min\{m, n\}$, there are no larger minors to check.

Rank of a 2×2 matrix

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then

$$\text{rank}(A) = \begin{cases} 2, & ad - bc \neq 0, \\ 1, & ad - bc = 0 \text{ and } A \neq 0, \\ 0, & A = 0. \end{cases}$$

Exercise Classify

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Solution Their ranks are 2, 1 and 0, respectively.

Rank of a 3×3 matrix

- ▶ If $\det A \neq 0$, then $\text{rank}(A) = 3$.
- ▶ If $\det A = 0$, inspect the 2×2 minors.
- ▶ One non-zero 2×2 minor gives rank 2.
- ▶ If all 2×2 minors vanish and $A \neq 0$, the rank is 1.
- ▶ If $A = 0$, the rank is 0.

Strategy

Start with the largest possible minors and move downward when necessary.

Exercise: compute rank with minors

Exercise Determine the rank of

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 0 \end{pmatrix}.$$

1. Explain why $\det A = 0$ without expanding it.
2. Find one non-zero 2×2 minor.
3. Combine the two facts to determine the rank.

Exercise: solution

Solution Since $R_2 = 2R_1$, $\det A = 0$ and $\text{rank}(A) < 3$.

$$\det \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} = 1 - 2 = -1 \neq 0.$$

This minor uses rows 1, 3 and columns 1, 2, so $\text{rank}(A) \geq 2$.

$$\boxed{\text{rank}(A) = 2.}$$

A vanishing determinant alone would only tell us that the rank is below 3.

Row rank and column rank

Theorem

$$\text{row rank}(A) = \text{column rank}(A) = \text{rank}(A).$$

The rank is the maximum number of linearly independent rows, and also the maximum number of linearly independent columns.

Connection with minors

A non-zero minor of order r identifies r independent rows and r independent columns.

In particular, $\text{rank}(A^\top) = \text{rank}(A)$.

Full rank

A matrix $A \in \mathbb{R}^{m \times n}$ has **full rank** when

$$\text{rank}(A) = \min\{m, n\}.$$

- ▶ If $m \geq n$, full rank means that all n columns are independent.
- ▶ If $m \leq n$, full rank means that all m rows are independent.

For a square matrix,

$$\text{rank}(A) = n \iff \det A \neq 0.$$

Rank and elementary row operations

The following operations preserve rank:

$$R_i \leftrightarrow R_j, \quad R_i \mapsto cR_i \ (c \neq 0), \quad R_i \mapsto R_i + cR_j \ (i \neq j).$$

Why?

Each operation is reversible. It changes the rows used to represent the same row space, without changing its dimension.

For an elementary matrix E ,

$$\text{rank}(EA) = \text{rank}(A).$$

Why arbitrary non-zero rows cannot be counted

Consider

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 0 \end{pmatrix}.$$

It has three non-zero rows, but the second row is twice the first.

$$\text{rank}(A) = 2.$$

The condition matters

Counting non-zero rows gives the rank once the matrix is in **row echelon form**. For a general matrix, non-zero rows may be dependent.

Pivots and row echelon form

The **pivot** of a non-zero row is its first non-zero entry. A matrix is in **row echelon form** if:

1. all zero rows are below the non-zero rows;
2. each pivot is strictly to the right of the pivot in the row above;
3. all entries below each pivot are zero.

$$\begin{pmatrix} 2 & 1 & 0 & 4 \\ 0 & 3 & 5 & 1 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Pivots need not equal 1.

Rank by row reduction: the missing final step

Apply $R_2 \mapsto R_2 - 2R_1$ and $R_3 \mapsto R_3 - R_1$:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -3 \end{pmatrix}.$$

Now swap R_2 and R_3 to obtain echelon form:

$$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -3 \\ 0 & 0 & 0 \end{pmatrix}.$$

Two pivots, hence $\boxed{\text{rank}(A) = 2}$.

Why counting pivots gives the rank

The non-zero rows of an echelon matrix are independent. Their pivots occupy different columns. In a linear combination equal to zero, the first pivot column forces the coefficient of the first row to be zero. The next pivot column forces the next coefficient to be zero, and so on.

$$\text{rank}(A) = \text{number of pivots in an echelon form of } A.$$

Two equivalent methods

Minors reveal rank through determinants. Elimination reveals rank through pivots.

From a system to its augmented matrix

For $Ax = b$, retain both coefficients and constants:

$$\begin{cases} x + y = 3, \\ 2x - y = 0 \end{cases} \longleftrightarrow [A \mid b] = \left(\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & -1 & 0 \end{array} \right).$$

One row is one equation

A row operation on $[A \mid b]$ acts on the entire equation, including its right-hand side.

We seek the same solutions using simpler equations.

Equivalent systems

Two systems are **equivalent** if they have the same solution set. Elementary row operations on the augmented matrix produce equivalent systems:

- ▶ exchange two equations;
- ▶ multiply an equation by a non-zero constant;
- ▶ add a multiple of another equation to an equation.

Reversibility preserves solutions

Every solution satisfies the transformed equations, and the inverse operations recover the original equations.

Gaussian elimination: the procedure

1. Start with the augmented matrix $[A \mid b]$.
2. Find a non-zero entry in the next available coefficient column. Swap rows if necessary.
3. Use this pivot to eliminate all entries below it.
4. Move to the remaining rows and columns. Skip columns with no available pivot.
5. Continue until echelon form is reached.

After elimination

Check for contradictions, compare ranks, and use back substitution when the solution is unique.

A 2×2 system: elimination

Exercise Solve

$$\begin{cases} x + y = 3, \\ 2x - y = 0. \end{cases}$$

Solution With $R_2 \mapsto R_2 - 2R_1$,

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & -1 & 0 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -3 & -6 \end{array} \right).$$

The second equation now involves only y .

A 2×2 system: back substitution

The echelon system is

$$\begin{cases} x + y = 3, \\ -3y = -6. \end{cases}$$

Read from bottom to top:

$$y = 2, \quad x = 3 - y = 1.$$

$$\boxed{(x, y) = (1, 2).}$$

Check in the original system: $1 + 2 = 3$ and $2(1) - 2 = 0$. Two coefficient pivots determine both unknowns.

When the first possible pivot is zero

Exercise Solve $2y = 4$, $x + y = 3$.

$$\left(\begin{array}{cc|c} 0 & 2 & 4 \\ 1 & 1 & 3 \end{array} \right) \xrightarrow[\sim]{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 2 & 4 \end{array} \right).$$

Solution Back substitution gives $y = 2$ and $x = 1$.

A zero entry does not end the algorithm

Look below it for a non-zero entry. If the whole available column is zero, move to the next column.

Compatible and incompatible systems

A system is **compatible** (or consistent) if it has at least one solution. It is **incompatible** (or inconsistent) if it has no solution. After elimination, interpret rows carefully:

$$(0 \ 0 \mid 0) \iff 0 = 0,$$

$$(0 \ 0 \mid 5) \iff 0 = 5.$$

The first row imposes no condition. The second makes the entire system impossible.

An incompatible 2×2 system

Consider

$$\begin{cases} x + y = 2, \\ 2x + 2y = 5. \end{cases}$$

With $R_2 \mapsto R_2 - 2R_1$,

$$\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & 2 & 5 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 1 \end{array} \right).$$

The last row is $0 = 1$. Thus the system has **no solution**.

$$\text{rank}(A) = 1, \quad \text{rank}([A \mid b]) = 2.$$

A compatible 2×2 system with repeated information

Now consider

$$\begin{cases} x + y = 2, \\ 2x + 2y = 4. \end{cases}$$

$$\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & 2 & 4 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 0 \end{array} \right).$$

Only $x + y = 2$ remains. For example, $(0, 2)$, $(1, 1)$ and $(2, 0)$ satisfy it.

$$\text{rank}(A) = \text{rank}([A \mid b]) = 1 < 2.$$

There are infinitely many points on this line.

Comparing the two ranks

Adding the column b cannot decrease rank and can increase it by at most one:

$$\text{rank}(A) \leq \text{rank}([A \mid b]) \leq \text{rank}(A) + 1.$$

- ▶ If b is a linear combination of the columns of A , it adds no independent column.
- ▶ If b is outside their span, it adds one independent column.

The equation $Ax = b$

Solving the system means expressing b as a linear combination of the columns of A .

The Rouché–Capelli theorem

Theorem (Rouché–Capelli) For $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, the system $Ax = b$ is compatible if and only if

$$\text{rank}(A) = \text{rank}([A \mid b]).$$

If the common rank is r :

- ▶ $r = n$: there is exactly one solution;
- ▶ $r < n$: there are infinitely many solutions.

The number to compare with rank

n is the number of **unknowns**, not the number of equations.

Why the Rouché–Capelli theorem works

Write $A = (C_1 \ \cdots \ C_n)$. Then

$$Ax = x_1 C_1 + \cdots + x_n C_n.$$

Thus $Ax = b$ has a solution exactly when $b \in \text{span}\{C_1, \dots, C_n\}$. This happens exactly when adding b leaves the column rank unchanged.

Uniqueness

If $r = n$, the columns are independent, so their coefficients representing b are unique. If $r < n$, a dependence relation lets us vary a solution and obtain infinitely many others.

Classification at a glance

Rank comparison	Conclusion
$\text{rank}(A) < \text{rank}([A \mid b])$	No solution
$\text{rank}(A) = \text{rank}([A \mid b]) = n$	Exactly one solution
$\text{rank}(A) = \text{rank}([A \mid b]) < n$	Infinitely many solutions

Order of the checks

First check compatibility. Only then decide whether a compatible system has a unique solution.

A 3×2 system: elimination

Exercise Classify and solve

$$\begin{cases} x + y = 2, \\ 2x - y = 1, \\ 3x = 3. \end{cases}$$

With $R_2 \mapsto R_2 - 2R_1$, $R_3 \mapsto R_3 - 3R_1$:

$$\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 3 & 0 & 3 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{array} \right).$$

One more step is needed to reach echelon form.

A 3×2 system: classification and solution

Apply $R_3 \mapsto R_3 - R_2$:

$$\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{array} \right).$$

Both ranks are 2, and $n = 2$: the solution is unique.

$$y = 1, \quad x = 2 - y = 1, \quad \boxed{(x, y) = (1, 1)}.$$

The third original equation gives $3x = 3$, as required. An extra equation can repeat information without causing a contradiction.

Changing one constant can destroy compatibility

Replace the third equation by $3x = 4$:

$$\begin{cases} x + y = 2, \\ 2x - y = 1, \\ 3x = 4. \end{cases} \quad \sim \quad \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & -3 & -3 \\ 0 & 0 & 1 \end{array} \right).$$

The same elimination produces the contradiction $0 = 1$.

$$\text{rank}(A) = 2, \quad \text{rank}([A \mid b]) = 3.$$

The system is incompatible.

A 2×3 system: classification today

Consider

$$\begin{cases} x + y + z = 2, \\ 2x - y + z = 1. \end{cases}$$

Apply $R_2 \mapsto R_2 - 2R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & -1 & 1 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & -3 & -1 & -3 \end{array} \right).$$

Both ranks are $2 < n = 3$: **infinitely many solutions**. We will describe the whole solution set using a parameter in Lecture 8.

Can a 2×3 system have a unique solution?

For two equations and three unknowns,

$$\text{rank}(A) \leq 2 < 3 = n.$$

Therefore a compatible system always has infinitely many solutions. An underdetermined system can also be incompatible. For example,

$$\begin{cases} x + y + z = 1, \\ 2x + 2y + 2z = 3 \end{cases} \implies 0 = 1.$$

Here the ranks are 1 and 2.

A 3×3 system: the problem

Exercise Solve by Gaussian elimination:

$$\begin{cases} x + y + z = 6, \\ 2x - y + z = 3, \\ x + 2y + 3z = 14. \end{cases}$$

1. Write the augmented matrix and reduce it to echelon form.
2. Compare ranks and classify the system.
3. Use back substitution and check the result.

A 3×3 system: first elimination step

Use $R_2 \mapsto R_2 - 2R_1$ and $R_3 \mapsto R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & 3 & 14 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 1 & 2 & 8 \end{array} \right).$$

For simpler arithmetic, swap R_2 and R_3 :

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & -3 & -1 & -9 \end{array} \right).$$

A 3×3 system: second elimination step

Use $R_3 \mapsto R_3 + 3R_2$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & -3 & -1 & -9 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 5 & 15 \end{array} \right).$$

There are three coefficient pivots and no contradiction:

$$\text{rank}(A) = \text{rank}([A \mid b]) = 3 = n.$$

There is exactly one solution.

A 3×3 system: back substitution and check

Solution The triangular system gives

$$5z = 15 \Rightarrow z = 3,$$

$$y + 2z = 8 \Rightarrow y = 8 - 6 = 2,$$

$$x + y + z = 6 \Rightarrow x = 6 - 2 - 3 = 1.$$

$$\boxed{(x, y, z) = (1, 2, 3).}$$

Check: $1 + 2 + 3 = 6$, $2 - 2 + 3 = 3$, $1 + 4 + 9 = 14$.

A singular 3×3 system: determinant is not enough

Consider

$$\begin{cases} x + y + z = 3, \\ 2x + 2y + 2z = 6, \\ x - y + z = 1. \end{cases}$$

Since $R_2 = 2R_1$ in the coefficient matrix, $\det A = 0$. After elimination and a row swap,

$$[A \mid b] \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & -2 & 0 & -2 \\ 0 & 0 & 0 & 0 \end{array} \right).$$

Both ranks are $2 < 3$: infinitely many solutions. Changing the second constant from 6 to 7 would produce $0 = 1$.